

SHENYANG, China

WMO: 543420

Lat: 41.7325N

Lon: 123.5099E

Elev: 161

StdP: 14.61

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.0	(c) -4.8	(d) -19.2	(e) 1.9	(f) 1.1	(g) -15.4	(h) 2.4	(i) 1.8	(j) 19.2	(k) 19.4	(l) 16.6	(m) 20.3	(n) 3.1	(o) 70	(p) 0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	89.4	74.0	87.3	72.9	85.4	72.0	78.4	85.4	77.0	83.4	75.6	81.6	8.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.4	138.4	82.6	75.0	131.9	81.0	73.6	126.0	79.8	42.2	85.3	40.7	83.7	39.3	81.8	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.9	(b) 16.7	(c) 14.3	(e) -13.5	(f) 92.6	(g) 5.3	(h) 2.7	(i) -17.3	(j) 94.5	(k) -20.4	(l) 96.1	(m) -23.4	(n) 97.6	(o) -27.3	(p) 99.5		
(a) 19.9	(b) 16.7	(c) 14.3	(e) -13.9	(f) 80.4	(g) 5.2	(h) 1.8	(i) -17.7	(j) 81.7	(k) -20.8	(l) 82.8	(m) -23.7	(n) 83.8	(o) -27.5	(p) 85.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	11.7	20.7	34.8	50.9	63.5	71.9	76.9	74.9	64.6	50.1	32.6	17.2		
	DBStd	23.77	8.50	9.66	8.68	8.08	6.17	4.62	3.60	4.45	5.84	8.10	10.10	9.38		
	HDD50	4217	1187	820	475	85	1	0	0	0	1	101	529	1018		
	HDD65	7357	1652	1240	936	425	101	5	0	2	76	464	973	1483		
	CDD50	3351	0	0	4	113	420	658	835	773	439	103	6	0		
	CDD65	1019	0	0	0	3	56	213	370	310	65	2	0	0		
	CDH74	8502	0	0	0	47	593	1716	3157	2455	519	15	0	0		
	CDH80	2439	0	0	0	3	125	511	1012	720	67	1	0	0		
	WSAvg	5.9	5.0	5.8	7.0	8.0	7.3	5.8	5.1	4.8	5.1	5.7	6.1	5.2		
	PrecAvg	26.6	0.3	0.4	0.6	1.6	2.3	3.5	6.7	6.2	2.6	1.7	0.7	0.4		
(15) Precipitation	PrecMax	38.2	1.1	1.5	2.8	4.1	6.9	7.8	14.3	14.1	7.4	5.6	2.9	1.3		
	PrecMin	10.0	0.0	0.0	0.0	0.0	0.0	1.0	1.4	0.5	0.4	0.0	0.0	0.0		
	PrecStd	5.6	0.2	0.4	0.6	0.9	1.3	1.6	3.3	3.5	1.4	1.1	0.5	0.4		
	PrecStd	5.6	0.2	0.4	0.6	0.9	1.3	1.6	3.3	3.5	1.4	1.1	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.2	49.5	64.5	79.4	87.2	91.8	92.6	91.2	85.0	76.0	63.0	44.8		
		MCWB	32.6	38.7	49.7	57.3	64.4	68.8	76.4	76.3	70.2	61.2	52.1	39.8		
	2%	DB	33.9	43.9	58.1	74.0	83.4	88.0	89.8	88.2	81.8	71.6	57.2	39.5		
		MCWB	28.9	35.6	44.8	55.2	62.3	68.0	76.1	75.5	67.0	58.9	48.9	34.6		
	5%	DB	30.4	40.3	53.7	70.3	80.4	85.6	87.6	86.3	79.6	67.8	53.1	36.0		
		MCWB	26.0	33.2	42.2	52.8	60.3	67.5	75.1	74.4	65.3	56.8	45.9	31.7		
	10%	DB	26.7	36.5	49.4	66.2	77.2	82.8	85.5	84.3	77.1	64.2	48.7	32.5		
		MCWB	22.8	30.6	39.3	50.2	58.6	67.1	74.1	73.4	64.1	54.2	42.6	28.5		
	0.4%	WB	34.0	40.4	50.8	61.6	68.0	73.8	80.5	80.1	73.9	64.0	53.6	40.9		
		MCDB	37.4	46.4	62.9	73.1	83.1	84.2	88.2	86.6	81.0	72.8	60.2	44.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	29.3	36.6	46.5	57.2	65.0	72.1	78.8	78.4	71.1	60.8	49.9	35.3		
		MCDB	33.1	42.7	56.0	70.7	78.3	82.0	86.7	84.8	77.0	68.1	55.9	38.7		
	5%	WB	26.2	33.7	43.0	54.5	62.8	70.7	77.3	77.1	68.7	58.5	46.8	31.9		
		MCDB	30.1	39.6	52.4	67.3	75.4	80.2	84.1	83.4	75.5	65.9	52.2	35.6		
	10%	WB	22.9	30.7	40.0	51.9	61.1	69.4	76.0	75.7	66.6	56.1	43.1	28.6		
		MCDB	26.6	36.4	48.5	63.9	73.4	78.4	82.4	81.5	73.5	62.9	48.4	32.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.4	20.0	18.5	20.8	20.6	17.4	14.4	15.5	20.9	20.7	17.5	18.3		
		MCDBR	23.7	22.5	24.8	27.5	26.2	22.3	17.9	18.3	23.6	24.9	23.2	21.2		
		MCWBR	19.6	15.7	14.9	13.6	10.7	8.1	7.0	7.5	10.2	13.6	16.0	17.6		
		MCDBR	23.0	20.9	23.2	23.5	21.5	17.3	14.1	14.1	18.4	21.6	20.8	20.0		
	5% WB	MCWBR	19.7	15.4	14.9	12.8	10.1	7.4	6.8	7.2	9.8	12.9	15.9	17.3		
		MCWBR	19.7	15.4	14.9	12.8	10.1	7.4	6.8	7.2	9.8	12.9	15.9	17.3		
(40) Clear-Sky Solar Irradiance	taub		0.374	0.440	0.498	0.600	0.651	0.728	0.707	0.619	0.541	0.552	0.465	0.395		
	taud		2.013	1.866	1.802	1.650	1.601	1.527	1.604	1.761	1.865	1.759	1.891	1.993		
	Ebn at Noon		229	230	233	219	213	197	199	213	219	192	192	207		
	Edh at Noon		37	51	61	76	82	89	82	68	57	56	43	36		
(44) All-Sky Solar Radiation	RadAvg		748	1022	1367	1619	1837	1679	1536	1503	1399	1040	717	607		
	RadStd		48	67	88	104	141	165	123	120	82	67	69	44		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a) Station Only	(b) Regional (9 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(a) Station Only	(b) Regional (9 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page